Anish Aggarwal

github/anishagg17 | **linkedin/anishagg** | anishagg17@gmail.com | anishagg17.github.io | +44 7470717446

Experience

Meta London, United Kingdom

Software Engineer *July 2024 – Present*

- Saved **$18M+/year** in compute by designing and shipping SQL-style predicate pushdown for ML data access on Scribe, cutting network transfer **70% (4 TB/s)** and CPU **42%**; drove adoption across **10+ ML teams**.
- Architected a multi-threaded C++ data-access layer for Scribe (Meta's **100 TB/s** distributed logging system), improving read throughput **35%** via multi-tier caching and concurrency control.
- Designed a **PID-controlled overload-protection system** for CPU/memory budgeting during traffic spikes, gracefully shedding excess load and **eliminating resource-exhaustion (OOM) incidents** while preserving availability for high-priority traffic.
- Built an automated **performance-regression framework** running CPU/network/memory A/B tests at code-review time, blocking **30+ regressions** pre-production across **20+ engineers**.

Google Warsaw, Poland

Software Engineer *May 2023 – July 2024*

- Integrated **ScaNN** approximate nearest-neighbor search into a **multi-modal retrieval pipeline**, accelerating vector search over high-dimensional embeddings under production workloads.
- Designed and shipped the **cluster lifecycle-management UI** for GKE VMware Admin Clusters in Google Cloud Console, replacing CLI-only workflows with self-service provisioning and upgrades for enterprise admins.
- Conducted **10+** technical interviews and mentored an intern through production delivery of a customer-facing feature.

Amazon Bengaluru, India

Software Engineer *Jan 2022 – May 2023*

- Designed a resilient backup payment flow that cut payment failures **48%**, recovering **$150K+** in the US region within 2 months of launch.
- Profiled and eliminated redundant network hops across payment services, cutting **P99 latency by 20ms** on **1M+ daily transactions**.
- Implemented asynchronous retry and fault-tolerance mechanisms, auto-recovering **34%** of transient payment failures.
- Led migration of **2,000+** dependencies across core payment services, remediating critical security vulnerabilities and unblocking long-stalled framework upgrades.

Elastic (Google Summer of Code) Remote

Open-Source Developer *May 2020 – Aug 2020*

- Implemented a text-diff utility for Elastic UI using the Myers difference algorithm (**sub-50ms** diffs on large documents).
- Designed and implemented an interactive component playground that streamlined component development for Elastic UI contributors.

Education

National Institute of Technology Hamirpur Hamirpur, India

B.Tech in Computer Science and Engineering ***(CGPA: 9.01/10)*** *2018 – 2022*

Coursework: Distributed Systems, Database Design, Operating Systems, Networks, Algorithms.

Achievements

Elastic Contributor Program: Ranked **#1 contributor worldwide**; awarded personally by the Elastic CEO.

Open Source: Contributor to Swift-NIO (Apple), Slate, and Susi.ai (FOSSASIA).

Skills

Languages: C++, Python, Java, Typescript, SQL, GraphQL
Systems: Distributed Systems, Kubernetes, Docker, GCP, AWS, Memcache, Elasticsearch
Performance: Linux internals, CPU/memory/network profiling, concurrency, PyTorch